05/05/26

# Pair of Linear equation in 2 variables

2 Burger + 1 coldrink = ₹100

$2x+y=100$

| Cost of 1 Burger | Cost of 1 glass drink |
|---|---|
| 25 | 50 |
| 35 | 30 |
| 50 | 0 |
| 30 | 40 |
| 1 | 98 |
| 40 | 20 |

$x+2y=80$

1 Burger + 2 Cold drink = ₹80

| 1 Burger | 1 glass cold drink |
|---|---|
| 0 | 40 |
| 80 | 0 |
| 30 | 25 |
| 70 | 5 |
| 79 | 0.5 |
| 40 | 20 |

General form of pair of linear equations in 2 variable

$$\begin{cases} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{cases}$$

# PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

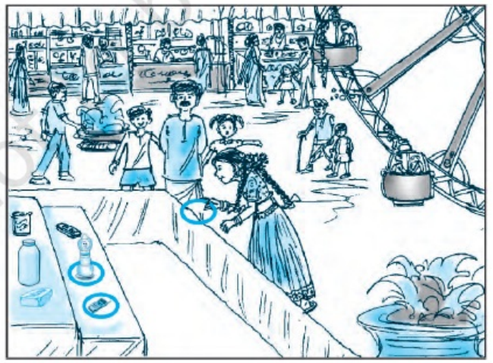
# 3

## 3.1 Introduction

You must have come across situations like the one given below :

Akhila went to a fair in her village. She wanted to enjoy rides on the Giant Wheel and play Hoopla (a game in which you throw a ring on the items kept in a stall, and if the ring covers any object completely, you get it). The number of times she played Hoopla is half the number of rides she had on the Giant Wheel. If each ride costs ₹ 3, and a game of Hoopla costs ₹ 4, how would you find out the number of rides she had and how many times she played Hoopla, provided she spent ₹ 20.

May be you will try it by considering different cases. If she has one ride, is it possible? Is it possible to have two rides? And so on. Or you may use the knowledge of Class IX, to represent such situations as linear equations in two variables.



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Let she ride $x$ times on Giant wheel & played hoopla $y$ times.

$3x + 4y = 20$ (i)

| $x$ | 4 | 0 |
|---|---|---|
| $y$ | 2 | 5 |

No. of times she played hoopla $= \frac{1}{2}$ No. of times she rides on wheel.

$y = \frac{1}{2}(x)$

$y = \frac{x}{2}$

$2y = x$

$x - 2y = 0$ (ii)

| $x$ | 0 | 2 |
|---|---|---|
| $y$ | 3 | 1 |

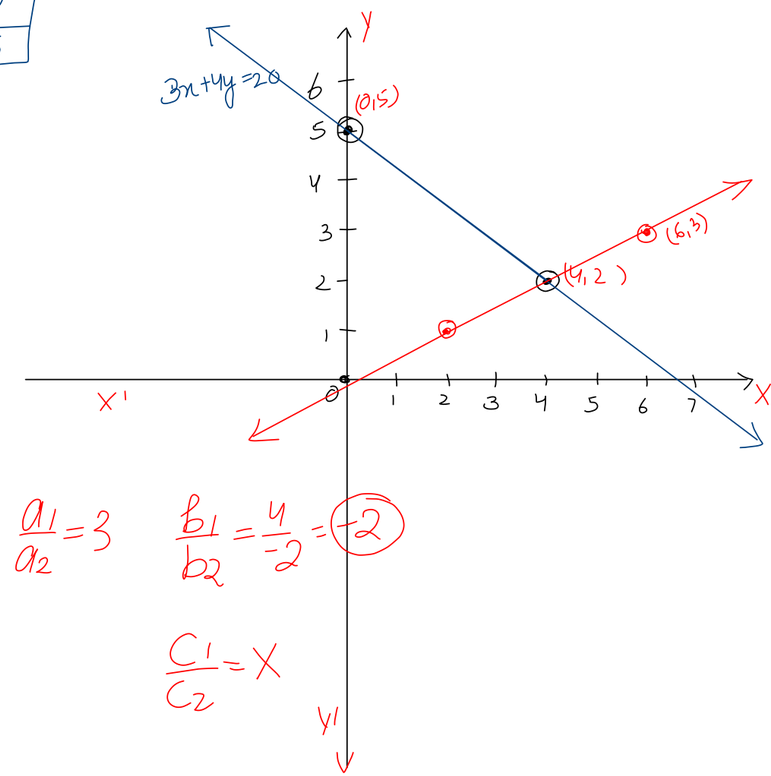
$3x + 4y + (-20) = 0$

$x + (-2y) + 0 = 0$

$a_1 = 3 \quad b_1 = 4 \quad c_1 = -20$

$a_2 = 1 \quad b_2 = -2 \quad c_2 = 0$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$




$\frac{a_1}{a_2} = 3 \quad \frac{b_1}{b_2} = \frac{4}{-2} = -2$

$\frac{c_1}{c_2} = \infty$

## 3.2 Graphical Method of Solution of a Pair of Linear Equations

A pair of linear equations which has no solution, is called an *inconsistent* pair of linear equations. A pair of linear equations in two variables, which has a solution, is called a *consistent* pair of linear equations. A pair of linear equations which are equivalent has infinitely many distinct common solutions. Such a pair is called a *dependent* pair of linear equations in two variables. Note that a dependent pair of linear equations is always consistent.

We can now summarise the behaviour of lines representing a pair of linear equations in two variables and the existence of solutions as follows:

(i) the lines may intersect in a single point. In this case, the pair of equations has a unique solution (consistent pair of equations).

(ii) the lines may be parallel. In this case, the equations have no solution (inconsistent pair of equations).

(iii) the lines may be coincident. In this case, the equations have infinitely many solutions [dependent (consistent) pair of equations].

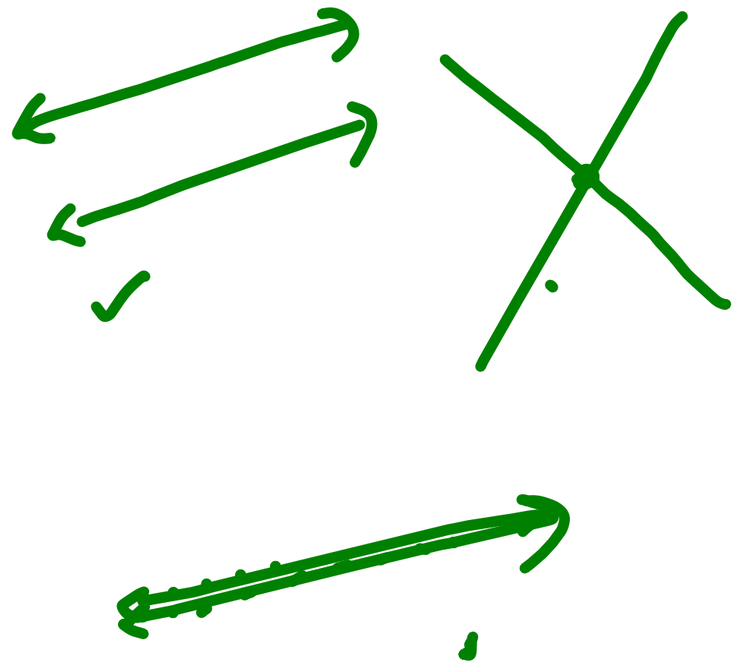
Consider the following three pairs of equations.

(i) $x - 2y = 0$ and $3x + 4y - 20 = 0$ (The lines intersect)

(ii) $2x + 3y - 9 = 0$ and $4x + 6y - 18 = 0$ (The lines coincide)

(iii) $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$ (The lines are parallel)

Let us now write down, and compare, the values of $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$ in all the three examples. Here, $a_1$, $b_1$, $c_1$ and $a_2$, $b_2$, $c_2$ denote the coefficents of equations given in the general form in Section 3.2.



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$2x+3y=9$

| $x$ | 1 | 3 |
|---|---|---|
| $y$ | 3 | 1 |

$4x+6y=18$

| $x$ | 6 | 4.5 |
|---|---|---|
| $y$ | $-1$ | 0 |

$$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$$

$2(1)+3y=9$

$2+3y=9$

$3y=7$

$y=\frac{7}{3}$

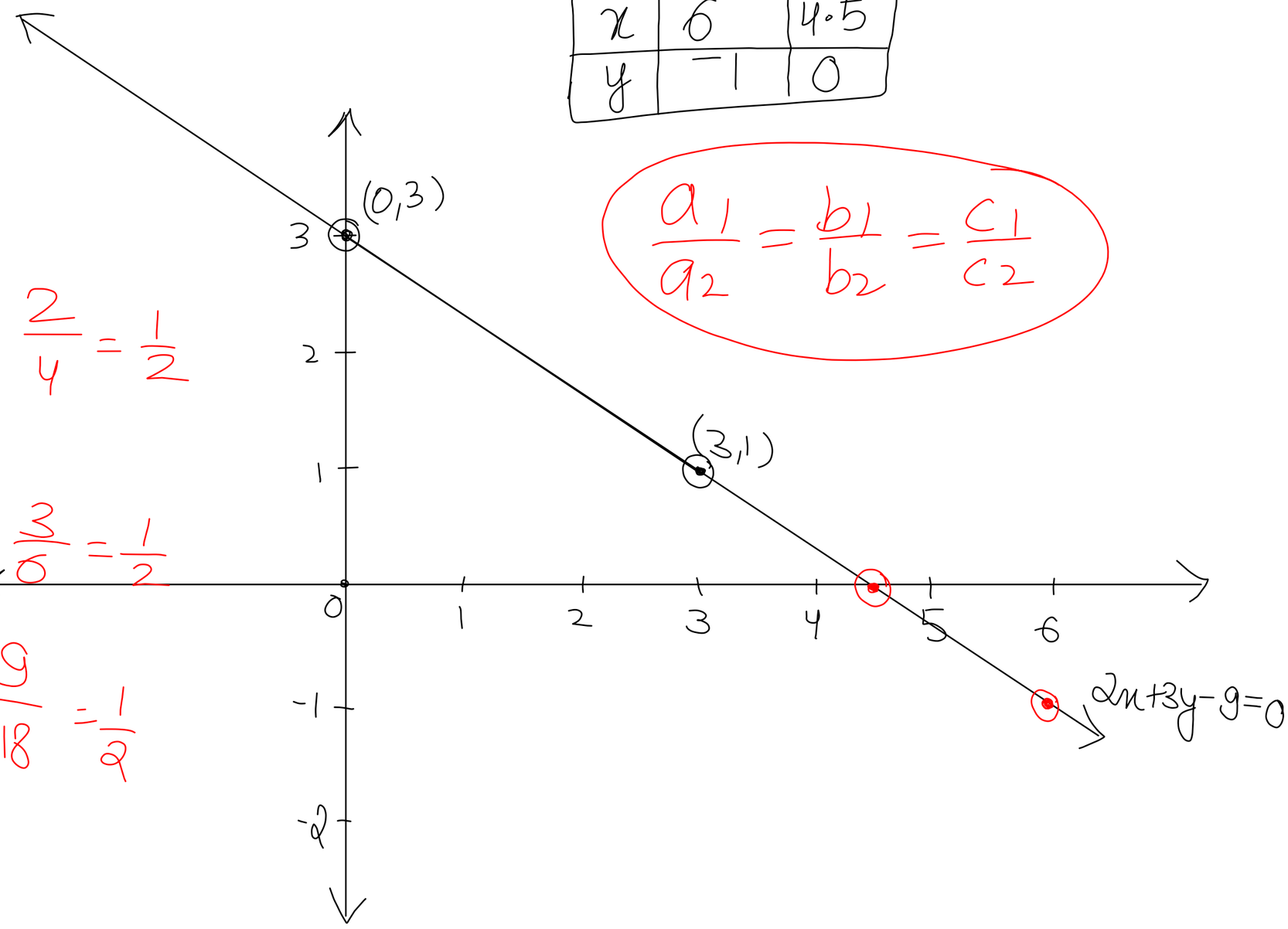
$$\frac{a_1}{a_2}=\frac{2}{4}=\frac{1}{2}$$

$$\frac{b_1}{b_2}=\frac{3}{6}=\frac{1}{2}$$

$$\frac{c_1}{c_2}=\frac{+9}{+18}=\frac{1}{2}$$

# $2(0)+3y=9$

$y=3$



$$\begin{cases} x+2y=4 \\ 2x+4y=12 \end{cases}$$

| $x$ | 0 | 2 |
|---|---|---|
| $y$ | 2 | 1 |

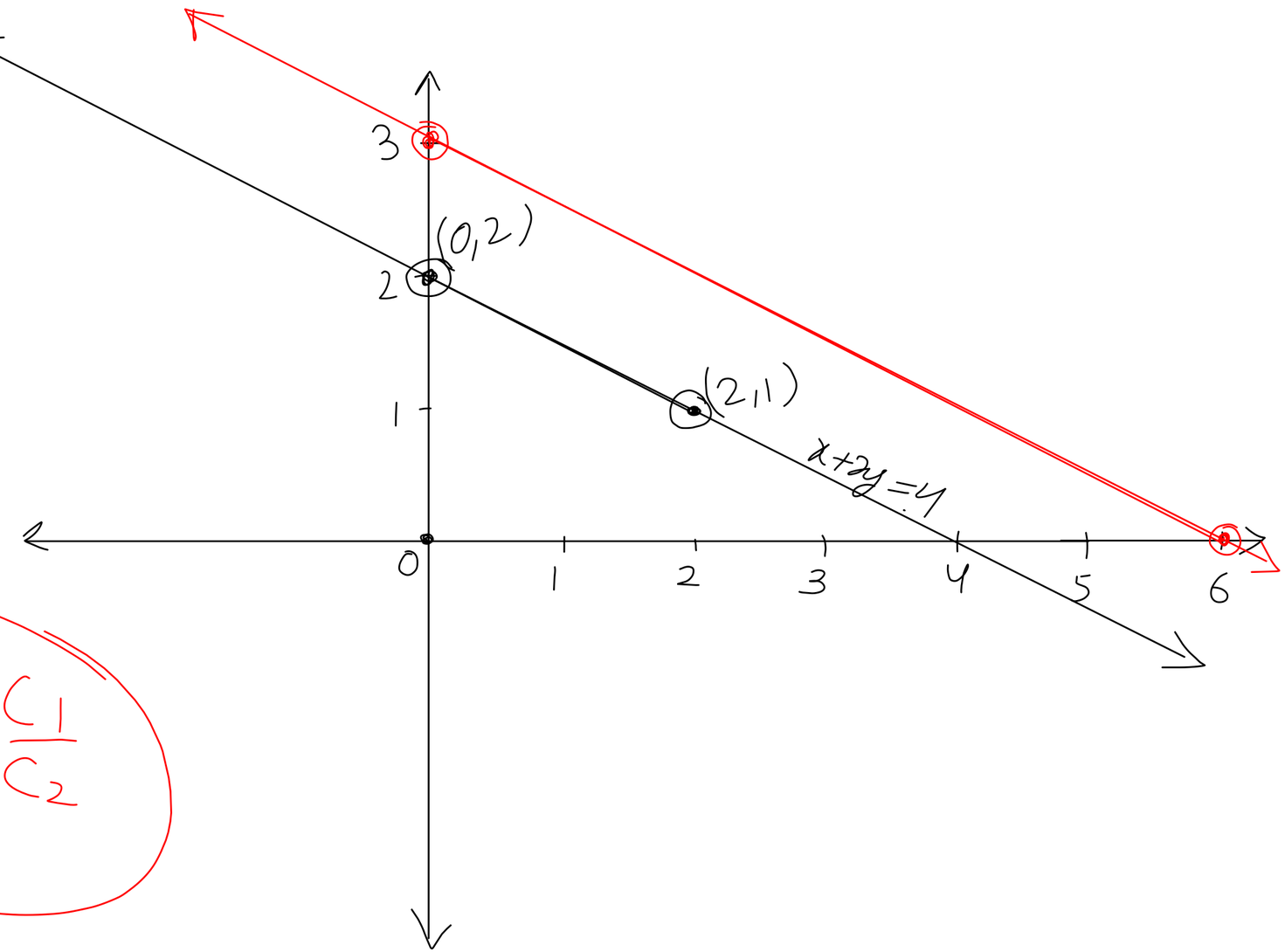
| $x$ | 6 | 0 |
|---|---|---|
| $y$ | 0 | 3 |

$$\frac{a_1}{a_2} = \frac{1}{2}$$

$$\frac{b_1}{b_2} = \frac{1}{2}$$

$$\frac{c_1}{c_2} = \frac{1}{3}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$



07/05/26

**Table 3.1**

| Sl No. | Pair of lines | $\frac{a_1}{a_2}$ | $\frac{b_1}{b_2}$ | $\frac{c_1}{c_2}$ | Compare the ratios | Graphical representation | Algebraic interpretation |
|---|---|---|---|---|---|---|---|
| 1. | $x-2y=0$<br>$3x+4y-20=0$ | $\frac{1}{3}$ | $\frac{-2}{4}$ | $\frac{0}{-20}$ | $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ | Intersecting lines | Exactly one solution (unique) → Consistent |
| 2. | $2x+3y-9=0$<br>$4x+6y-18=0$ | $\frac{2}{4}$ | $\frac{3}{6}$ | $\frac{-9}{-18}$ | $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ | Coincident lines | Infinitely many solutions → Consistent |
| 3. | $x+2y-4=0$<br>$2x+4y-12=0$ | $\frac{1}{2}$ | $\frac{2}{4}$ | $\frac{-4}{-12}$ | $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ | Parallel lines | No solution → Inconsistent |

From the table above, you can observe that if the lines represented by the equation

$$\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$$

System of linear equations in 2 variables

and

are (i) intersecting, then $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

(ii) coincident, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(iii) parallel, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

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**Example 1 :** Check graphically whether the pair of equations

$$x + 3y = 6 \quad (1)$$

and

$$2x - 3y = 12 \quad (2)$$

is consistent. If so, solve them graphically.

$\frac{a_1}{a_2} = \frac{1}{2}$

$\frac{b_1}{b_2} = -1$

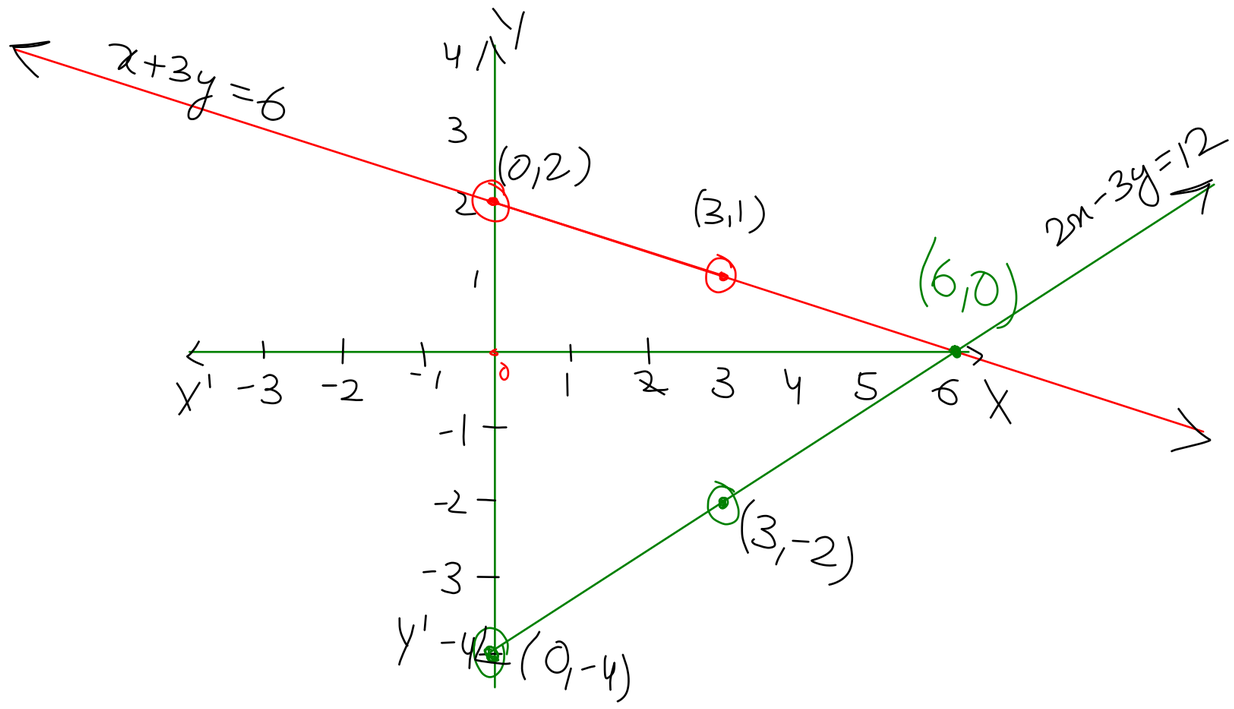
$\boxed{\frac{a_1}{a_2} \neq \frac{b_1}{b_2}}$ {The system is consistent & having unique solution}

$x+3y=6$

| $x$ | 0 | 3 |
|---|---|---|
| $y$ | 2 | 1 |

$2x-3y=12$

| $x$ | 0 | 3 |
|---|---|---|
| $y$ | -4 | -2 |




The solution is $x = 6$, $y = 0$

## EXERCISE 3.1

**1.** Form the pair of linear equations in the following problems, and find their solutions graphically.

(i) 10 students of Class X took part in a Mathematics quiz. If the number of girls is 4 more than the number of boys, find the number of boys and girls who took part in the quiz.

(ii) 5 pencils and 7 pens together cost ₹ 50, whereas 7 pencils and 5 pens together cost ₹ 46. Find the cost of one pencil and that of one pen.

$\frac{a_1}{a_2} = 1 \quad \frac{b_1}{b_2} = -1$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

$5x+7y=50$

$7x+5y=46$

Let the no. of girls $= x$

boys $= y$.

$x+y=10 \rightarrow x+y-10=0$ (i)

$x = 4+y$

$x-y-4=0$ — (ii)

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\frac{1}{1} -$

$5x+7y=50$

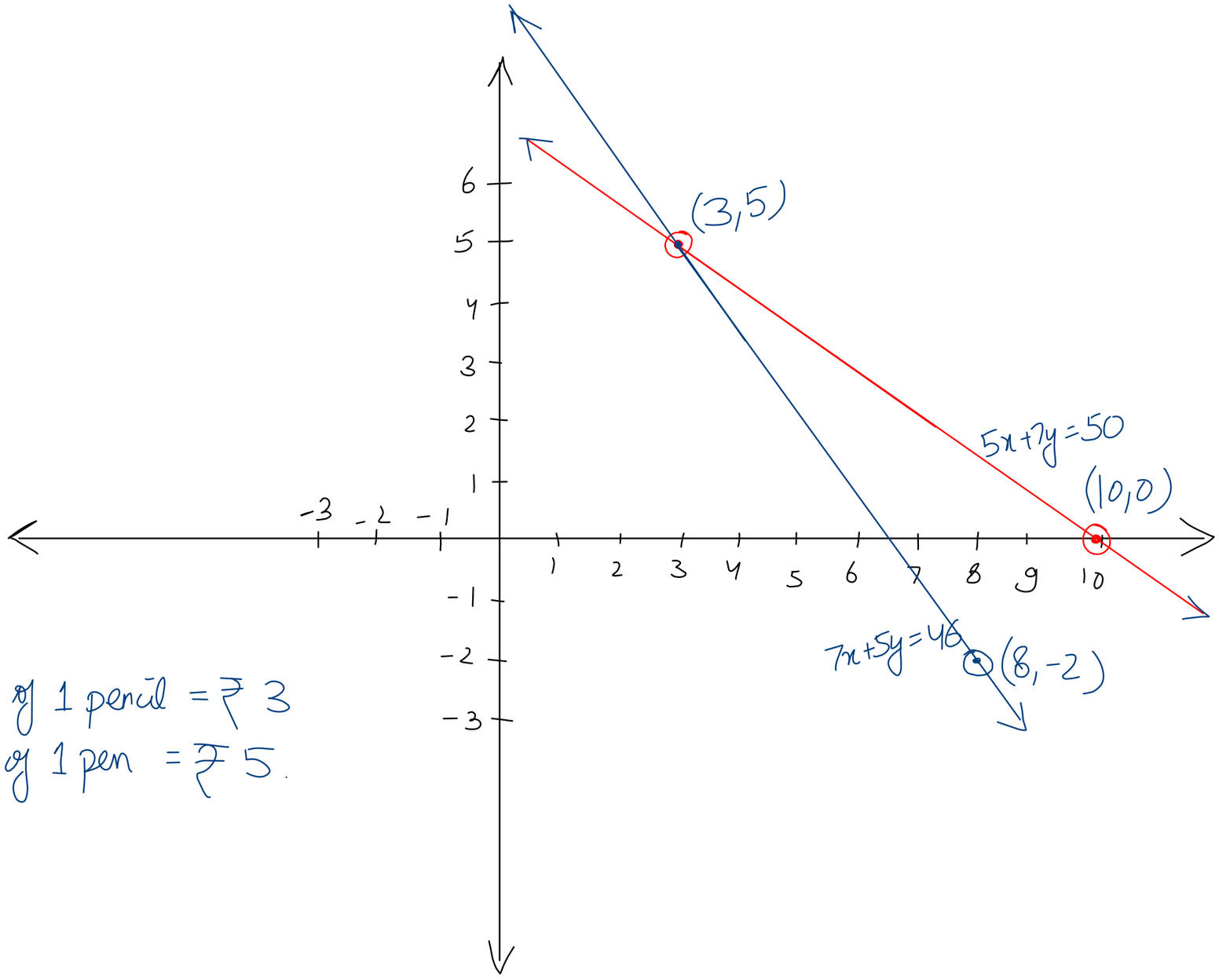
$7x+5y=46$

| $x$ | 10 | 3 |
|---|---|---|
| $y$ | 0 | 5 |

| $x$ | 3 | 8 |
|---|---|---|
| $y$ | 5 | $-2$ |

21
25

35
k5




Cost of 1 pencil = ₹ 3

Cost of 1 pen = ₹ 5.

**2.** On comparing the ratios $\frac{a_1}{a_2}, \frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$, find out whether the lines representing the following pairs of linear equations intersect at a point, are parallel or coincident:

(i) $5x-4y+8=0$
$7x+6y-9=0$

$\frac{a_1}{a_2}=\frac{5}{7}$

(ii) $9x+3y+12=0$
$18x+6y+24=0$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

(iii) $6x-3y+10=0$
$2x-y+9=0$

$\frac{b_1}{b_2}=\frac{-4}{6}=\frac{-2}{3}$

**3.** On comparing the ratios $\frac{a_1}{a_2}, \frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$, find out whether the following pair of linear equations are consistent, or inconsistent.

(i) $3x+2y=5$ ; $2x-3y=7$

(ii) $2x-3y=8$ ; $4x-6y=9$

(iii) $\frac{3}{2}x+\frac{5}{3}y=7$ ; $9x-10y=14$

(iv) $5x-3y=11$ ; $-10x+6y=-22$

(v) $\frac{4}{3}x+2y=8$ ; $2x+3y=12$

**4.** Which of the following pairs of linear equations are consistent/inconsistent? If consistent, obtain the solution graphically:

(i) $x+y=5$, $2x+2y=10$

(ii) $x-y=8$, $3x-3y=16$

(iii) $2x+y-6=0$, $4x-2y-4=0$

(iv) $2x-2y-2=0$, $4x-4y-5=0$

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**5.** Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden.

**6.** Given the linear equation $2x + 3y - 8 = 0$, write another linear equation in two variables such that the geometrical representation of the pair so formed is:

(i) intersecting lines (ii) parallel lines

(iii) coincident lines

**7.** Draw the graphs of the equations $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the $x$-axis, and shade the triangular region.

(5) Length → x m
Width → y m

$x + y = 36$
$x - y = 4$
$+ \quad - \quad =$
$2y = 32$
$y = 16$

$x - y = 4$
$x + y = 36$
$2x = 40$
$x = 20$

$L = 4 + w$
$x = 4 + y$
$x - y - 4 = 0$ (i)

$P = 2(l + w)$
$\frac{P}{2} = l + w$
$36 = x + y$
$x + y = 36$ (ii)
$x + y - 36 = 0$

# Algebraic method to find the solution of system of linear equation.

## ① Substitution Method

$$\begin{cases} 3x+4y=20 \\ x-2y=0 \end{cases} \Rightarrow$$

$x = 2y$

$3(2y)+4y=20$

$6y+4y=20$

$10y=20$

$y=2$

$x=2(2)$

$x=4$

$2y=x$

$y=\frac{x}{2}$

$3x+4\left(\frac{x}{2}\right)=20$

$3x+2x=20$

$5x=20$

$x=4$

$y=\frac{4}{2}$

$y=2$

## ② Elimination method

$$3x + 4y = 20$$
$$x - 2y = 0 \quad) \times 2 \longrightarrow \times 3$$
$$2x - 4y = 0$$

$$3x + 4y = 20$$
$$3x - 6y = 0$$
$$+$$
$$10y = 20$$
$$y = 2$$

$$3x + 4y = 20$$
$$2x - 4y = 0$$
$$+$$
$$5x = 20$$
$$x = 4$$

## II. AGE RELATED QUESTIONS

**1.** Ten years hence, a man's age will be twice the age of his son. Ten years ago, man was four times as old as his son. Find their present ages.

**2.** A man's age is three times the sum of the ages of his two sons. After 5 years his age will be twice the sum of the ages of his two sons. Find the age of the man.

Let present age of man is $x$.

Son is $y$.

After 10 years

Man's age will be $= x+10$

Son's age will be $= y+10$

Man's age $= 2 \times$ Son's age

$x+10 = 2(y+10)$

$x+10 = 2y+20$

$x-2y = 10$ —(i)

$x = 10+2y$

$x = 10+2(20)$

$x = 50$

10 years ago

Man's age will be $x-10$

Son's age will be $y-10$

Man's age $= 4 \times$ Son's age

$x-10 = 4(y-10)$

$x-10 = 4y-40$

$x-4y = -30$ —(ii)

$10+2y-4y = -30$

$10-2y = -30$

$-2y = -40$

$y = \frac{+40}{+2} = 20$

$y = 20$

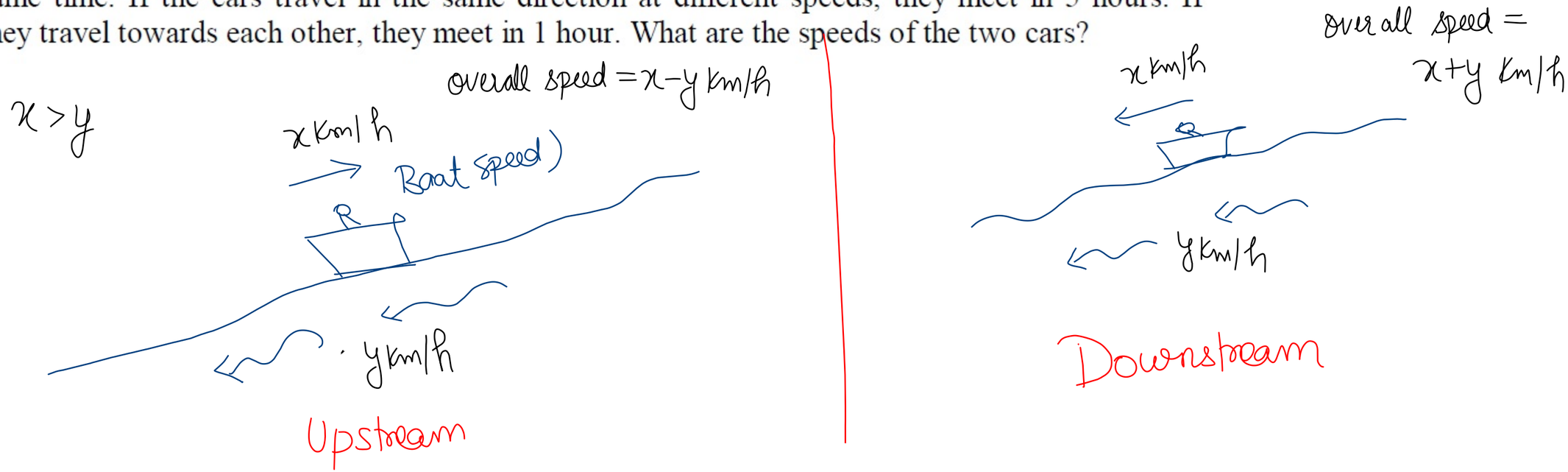
Present age of man = 50 years

Son's present age = 20 years

# III. SPEED, DISTANCE AND TIME RELATED QUESTIONS

**1.** A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km down-stream. Determine the speed of the stream and that of the boat in still water.

**2.** Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars?



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① Let the speed of boat is $x$ km/h
stream is $y$ km/h

Time = distance / speed

Time in 30 km upstream + Time in 44 km downstream = 10 hours

$$\frac{30}{x-y} + \frac{44}{x+y} = 10 \quad -(i)$$

40 km upstream + 55 km downstream = 13

$$\frac{40}{x-y} + \frac{55}{x+y} = 13 \quad -(ii)$$

$30\left(\frac{1}{x-y}\right) + 44\left(\frac{1}{x+y}\right) = 10$

$40\left(\frac{1}{x-y}\right) + 55\left(\frac{1}{x+y}\right) = 13$

Let $\frac{1}{x-y} = a$, $\frac{1}{x+y} = b$

$30a + 44b = 10$ ×4

$40a + 55b = 13$ ×3

$120a + 176b = 40$

$-\ 120a + 165b = 39$

$11b = 1$

$b = \frac{1}{11}$

$30a + \frac{44}{11} = 10$

$30a = 10 - \frac{44}{11} = \frac{46}{11}$

$30a = \frac{46}{11}$

$a = \frac{46}{30 \times 11} = \frac{23}{165}$

$\frac{1}{x-y} = \frac{23}{165}$, $\frac{1}{x+y} = \frac{1}{11}$

$x + y = 11$ ×23

$23x - 23y = 165$

$23x + 23y = 253$

$23x - 23y = 165$

$46x = 418$

$x = \frac{418}{46} = \frac{209}{23} = 9\frac{2}{23}$ km/h

$11 + y = 11$

$y = 3$

$161 + 23 = 184$

8 9 10 ~~11 12~~

9.00 − 7.43 = 1.57

$7.43 + y = 9$

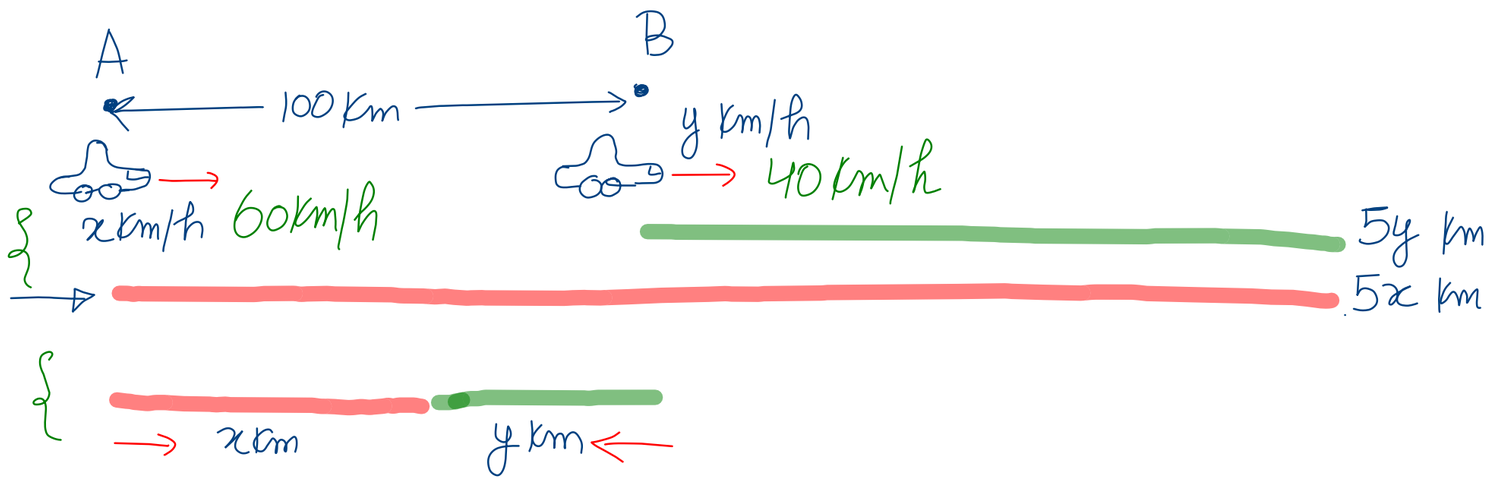
$y = 1.57$

$\frac{418}{46} = \frac{209}{23} = 9\frac{2}{23}$

## III. SPEED, DISTANCE AND TIME RELATED QUESTIONS

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$5x - 5y = 100$

$5(x-y) = 100$

$\boxed{x-y=20}$ — (i)

$x+y=100$ — (ii)

$2x = 120$

$\boxed{x=60}$ $y=40$

## IV. GEOMETRICAL FIGURES RELATED QUESTIONS

**1.** The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle.

**2.** The length of a room exceeds its breadth by 3 metres. If the length is increased by 3 metres and the breadth is decreased by 2 metres, the area remains the same. Find the length and the breadth of the room.

## FRACTION PROBLEMS

**1.** A fraction becomes $\frac{9}{11}$, if 2 is added to both the numerator and the denominator. If 3 is added to both the numerator and the denominator it becomes $\frac{5}{6}$. Find the fraction.

**2.** The sum of numerator and denominator of a fraction is 12. If the denominator is increased by 3 then the fraction becomes $\frac{1}{2}$. Find the fraction.

# TWO-DIGIT PROBLEMS

Original number = $10x+y$
Reverse number = $10y+x$

**1.** The sum of the digits of a two digit number is 12. The number obtained by interchanging the two digits exceeds the given number by 18. Find the number.

**2.** Seven times a two-digit number is equal to four times the number obtained by reversing the order of its digit. If the difference between the digits is 3, then find the number.

Let the two-digit no is $\overset{T}{x}\ \overset{O}{y}$ → 57

number after interchanging the digits will be $y\ x$

$xy = 10x+y$

$12 = 10+2$

$28 = 20+8$

New no = 18 + Original no.

$10y+x = 18+10x+y$

$9y-9x = 18$

$y-x = 2 \Rightarrow -x+y = 2$ —(ii)

$x+y = 12$ —(i)

$x+y = 12$
$-x+y = 2$
$2y = 14$
$y = 7$

$x+7 = 12$
$x = 5$

(iv) $\frac{2}{10}x + \frac{3}{10}y = \frac{13}{10}$

$\Rightarrow 2x+3y=13$ → $2x = 13-3y$

$4x+5y=23$ ← $x = \frac{13-3y}{2}$

$\Rightarrow x = \frac{13-9}{2}$

$= \frac{4}{2} = 2$

$x=2$

$\frac{2}{4}\left(\frac{13-3y}{2}\right)+5y=23$

$\Rightarrow 26-6y+5y=23$

$\Rightarrow -y=-3$

$y=3$

$x=2$
$y=3$ Solution

## EXERCISE 3.2

**1.** Solve the following pair of linear equations by the substitution method.

(i) $x+y=14$
$x-y=4$

(ii) $s-t=3$
$\frac{s}{3}+\frac{t}{2}=6$

(iii) $3x-y=3$
$9x-3y=9$

(iv) $0.2x+0.3y=1.3$
$0.4x+0.5y=2.3$

(v) $\sqrt{2}x+\sqrt{3}y=0$
$\sqrt{3}x-\sqrt{8}y=0$

(vi) $\frac{3x}{2}-\frac{5y}{3}=-2$
$\frac{x}{3}+\frac{y}{2}=\frac{13}{6}$

**2.** Solve $2x + 3y = 11$ and $2x - 4y = -24$ and hence find the value of '$m$' for which $y=mx+3$.

HW

$2x+3y=11$

$2x-4y=-24$

$7y=35$

$y=5$

$2x+15=11$

$2x=-4$

$x=-2$

$y=mx+3$

$5=m(-2)+3$

$2=-2m$

$-1=m$

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**3.** Form the pair of linear equations for the following problems and find their solution by substitution method.

(i) The difference between two numbers is 26 and one number is three times the other. Find them.

(ii) The larger of two supplementary angles exceeds the smaller by 18 degrees. Find them.

(iii) The coach of a cricket team buys 7 bats and 6 balls for ₹ 3800. Later, she buys 3 bats and 5 balls for ₹ 1750. Find the cost of each bat and each ball.

(iv) The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is ₹ 105 and for a journey of 15 km, the charge paid is ₹ 155. What are the fixed charges and the charge per km? How much does a person have to pay for travelling a distance of 25 km?

(v) A fraction becomes $\frac{9}{11}$, if 2 is added to both the numerator and the denominator. If, 3 is added to both the numerator and the denominator it becomes $\frac{5}{6}$. Find the fraction.

(iv) Let the fixed charge of taxi = $x$ Rs

Charge per Km = $y$ Rs/Km

$x + 10y = 105$

$x + 15y = 155$

$-5y = -50$

$y = 10$

$x + 100 = 105$

$x = 5$

Charge for 25 kms $= 5 + 10 \times 25$

$= 5 + 250$

$= ₹ 255$

# MCQ WORKSHEET-II
## CLASS X : CHAPTER - 3
## PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

1. If a pair of equation is consistent, then the lines will be
   (a) parallel (b) always coincident
   (c) always intersecting (d) intersecting or coincident

2. The solution of the equations x + y = 14 and x – y =4 is
   (a) x = 9 and y = 5 (b) x = 5 and y = 9 (c) x = 7 and y = 7 (d) x = 10 and y = 4

3. The sum of the numerator and denominator of a fraction is 12. If the denominator is increased by 3, the fraction becomes ½ , then the fraction
   (a) $\frac{4}{7}$ (b) $\frac{5}{7}$ (c) $\frac{6}{7}$ (d) $\frac{3}{7}$

4. The value of k for which the system of equations x – 2y = 3 and 3x + ky = 1 has a unique solution is
   (a) k = – 6 (b) k ≠ – 6 (c) k = 0 (d) no value

5. If a pair of equation is inconsistent, then the lines will be
   (a) parallel (b) always coincident
   (c) always intersecting (d) intersecting or coincident

$2x - y = 3$
$x + y = 12$
$3x = 15$
$x = 5$

$\frac{x}{y+3} = \frac{1}{2}$
$2x = y + 3$
$2x - y = 3$ -(ii)

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \frac{1}{3} \neq \frac{-2}{k}$
$k \neq -6$ ✓

6. The value of k for which the system of equations $2x + 3y = 5$ and $4x + ky = 10$ has infinite many solution is

(a) $k = -3$ (b) $k \neq -3$ (c) $k = 0$ (d) none of these

7. The value of k for which the system of equations $kx - y = 2$ and $6x - 2y = 3$ has a unique solution is

(a) $k = -3$ (b) $k \neq 3$ (c) $k = 0$ (d) $k \neq 0$

8. Sum of two numbers is 35 and their difference is 13, then the numbers are

(a) 24 and 12 (b) 24 and 11 (c) 12 and 11 (d) none of these

9. The solution of the equations $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$ is

(a) $x = 1$ and $y = 2$ (b) $x = 2$ and $y = 3$ (c) $x = 3$ and $y = 4$ (d) $x = 5$ and $y = 4$

10. The solution of the equations $x + 2y = 1.5$ and $2x + y = 1.5$ is

(a) $x = 1$ and $y = 1$ (b) $x = 1.5$ and $y = 1.5$ (c) $x = 0.5$ and $y = 0.5$ (d) none of these

11. The value of k for which the system of equations $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution is

(a) 10 (b) 6 (c) 3 (d) 1

12. The value of k for which the system of equations $3x + 5y = 0$ and $kx + 10y = 0$ has a non-zero solution is

(a) 0 (b) 2 (c) 6 (d) 8

**5.** If $2x + 3y = 0$ and $4x - 3y = 0$, then $x + y$ equals:

(a) 0 (b) $-1$ (c) 1 (d) 2

**6.** If $\sqrt{a}x - \sqrt{b}y = b - a$ and $\sqrt{b}x - \sqrt{a}y = 0$, then the value of x, y is:

(a) $a + b$ (b) $a - b$ (c) $\sqrt{ab}$ (d) $-\sqrt{ab}$

**7.** If $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, then $x + y$ equals:

(a) $\frac{1}{6}$ (b) $-\frac{1}{6}$ (c) $\frac{5}{6}$ (d) $-\frac{5}{6}$

**8.** If $31x + 43y = 117$ and $43x + 31y = 105$, then value of $x - y$ is:

(a) $\frac{1}{3}$ (b) $-3$ (c) 3 (d) $-\frac{1}{3}$

**9.** If $19x - 17y = 55$ and $17x - 19y = 53$, then the value of $x - y$ is:

(a) $\frac{1}{3}$ (b) $-3$ (c) 3 (d) 5

⑦ $\frac{1}{x} = a$ , $\frac{1}{y} = b$

$(2a + 3b = 13) \times 5$

$(5a - 4b = -2) \times 2$

$10a + 15b = 65$

$10a - 8b = -4$

$23b = 69$

$b = \frac{69}{23} = 3$

$\frac{1}{y} = 3$ → $3y = 1$ → $y = \frac{1}{3}$

$b = 3$

$\frac{1}{x} = 2$ → $x = \frac{1}{2}$

$a = 2$

$x + y = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$

$\frac{a-b}{b-a} = \frac{a-b}{-(a-b)}$

⑤ $2x + 3y = 0$

$4x - 3y = 0$

$6x = 0$

$x = 0$

$2(0) + 3y = 0$

$3y = 0$

$y = 0$

⑥ $[\sqrt{a}x - \sqrt{b}y = b - a] \times \sqrt{b}$

$[\sqrt{b}x - \sqrt{a}y = 0] \times \sqrt{a}$

$\sqrt{ab}x - by = (b-a)\sqrt{b}$

$-\sqrt{ab}x + ay = 0$

$-by + ay = (b-a)\sqrt{b}$

$y(-b+a) = (b-a)\sqrt{b}$

$y(a-b) = -(a-b)\sqrt{b}$

$y = -\sqrt{b}$

$x \times y = +\sqrt{a} \times +\sqrt{b}$

$= \sqrt{ab}$

$\sqrt{a}x - \sqrt{b}(-\sqrt{b}) = b - a$

$\sqrt{a}x + b = b - a$

$x = \frac{-a}{\sqrt{a}}$

$x = -\sqrt{a}$

**5.** If $2x + 3y = 0$ and $4x - 3y = 0$, then $x + y$ equals:

(a) 0 (b) $-1$ (c) 1 (d) 2

**6.** If $\sqrt{a}x - \sqrt{b}y = b - a$ and $\sqrt{b}x - \sqrt{a}y = 0$, then the value of x, y is:

(a) $a + b$ (b) $a - b$ (c) $\sqrt{ab}$ (d) $-\sqrt{ab}$

**7.** If $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, then $x + y$ equals:

(a) $\frac{1}{6}$ (b) $-\frac{1}{6}$ (c) $\frac{5}{6}$ (d) $-\frac{5}{6}$

**8.** If $31x + 43y = 117$ and $43x + 31y = 105$, then value of $x - y$ is:

(a) $\frac{1}{3}$ (b) $-3$ (c) 3 (d) $-\frac{1}{3}$

**9.** If $19x - 17y = 55$ and $17x - 19y = 53$, then the value of $x - y$ is:

(a) $\frac{1}{3}$ (b) $-3$ (c) 3 (d) 5

$\frac{a-b}{b-a} = \frac{a-b}{-(a-b)}$

⑦ $\frac{1}{x} = a$ , $\frac{1}{y} = b$

$2a + 3b = 13$ ×5

$5a - 4b = -2$ ×2

$10a + 15b = 65$

$10a - 8b = -4$

$23b = 69$

$b = \frac{69}{23} = 3$

$\frac{1}{y} = 3$ → $y = \frac{1}{3}$

$3y = 1$

$\frac{1}{x} = 2$

$x = \frac{1}{2}$

$2x = 1$

$a = 2$

$b = 3$

$x + y = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$

⑧ $31x + 43y = 117$

$43x + 31y = 105$

$74x + 74y = 222$

$74(x + y) = 222$

$x + y = 3$ —(i)

74 ×3 222

$31x + 43y = 117$

$43x + 31y = 105$

$-12x + 12y = 12$

$12(-x + y) = 12$

$-x + y = 1$ —(ii)

$x + y = 3$

$-x + y = 1$

$2y = 4$

$y = 2$

$x + 2 = 3$

$x = 1$

**21.** $\frac{3}{x}-\frac{1}{y}+9=0;\quad \frac{2}{x}+\frac{3}{y}=5\quad (x\neq 0, y\neq 0)$

**22.** $2x-\frac{3}{y}=9;\quad 3x+\frac{7}{y}=2,\quad y\neq 0$

**23.** $x+\frac{6}{y}=6;\quad 3x-\frac{8}{y}=5,\quad y\neq 0$

**24.** $\frac{4}{x}+5y=7;\quad \frac{3}{x}+4y=5,\quad x\neq 0$

**25.** $\frac{x}{3}+\frac{y}{4}=11;\qquad \frac{5x}{6}-\frac{y}{3}+7=0$

**26.** $\frac{x+y-8}{2}=\frac{x+2y-14}{3}=\frac{3x+y-12}{11}$

**27.** $\frac{x+y}{xy}=2;\frac{x-y}{xy}=6,\ \ x\neq 0, y\neq 0.$

**28.** $\frac{xy}{x+y}=\frac{6}{5};\frac{xy}{y-x}=6; x+y\neq 0, y-x\neq 0.$